Project Design Phase
Solution Architecture

Date	15 February 2025
Team ID	SWTID-2026-8331
Project Name	Team Enrollment Confirmation – Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Solution Architecture:

The solution imports team enrolment data from a spreadsheet into ServiceNow using Import Sets and Transform Maps, validates it, and confirms enrolment automatically. The architecture bridges the business problem (slow, error-prone manual entry) and the technology solution. Its goals are to:

- Find the best tech solution to eliminate manual, error-prone data entry.
- Describe the structure, characteristics and behaviour of the solution to project stakeholders.
- Define features, development phases and solution requirements.
- Provide specifications according to which the solution is defined, managed and delivered.

Solution Architecture Diagram:

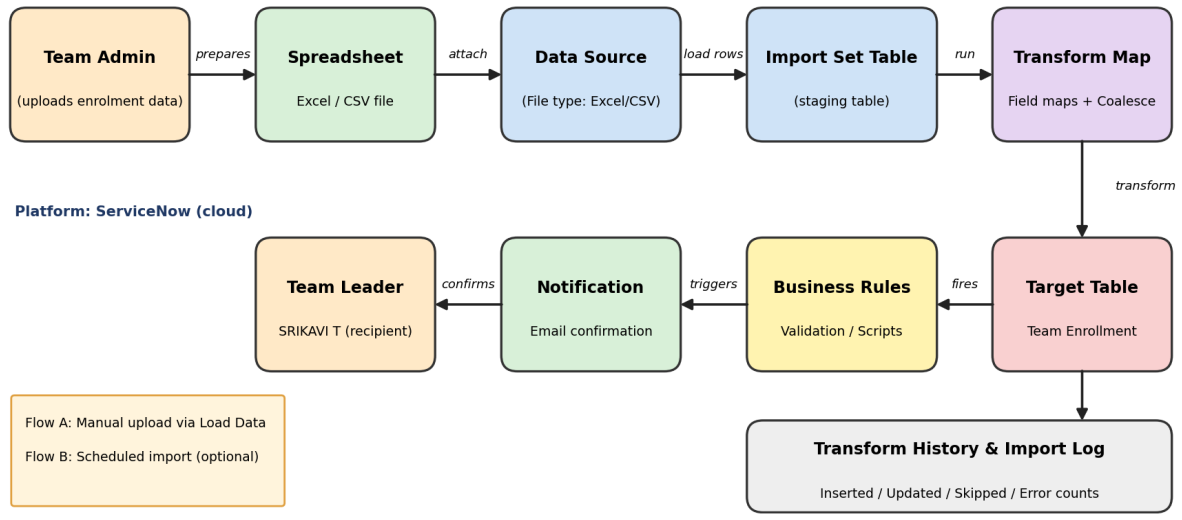


Figure 1: Architecture and data flow of the Team Enrollment import using Transform Maps

Components

S.No.	Component	Description
1	Team Admin / User	Prepares the enrolment spreadsheet (Team ID, team size, leader, members) and starts the import.
2	Spreadsheet (Excel/CSV)	Source file that holds the team enrolment records to be imported.
3	Data Source	ServiceNow record that points to the uploaded file and defines its format and header row.

S.No.	Component	Description
4	Import Set Table	Staging table where the raw spreadsheet rows are loaded before any changes are made.
5	Transform Map	Maps each staging column to a target field; Team ID is set as the coalesce field to prevent duplicates.
6	Business Rules / Transform Scripts	Validate and clean data (mandatory fields, trimming, team size check) during onBefore/onAfter events.
7	Target Table (Team Enrollment)	Final table storing confirmed enrolment records.
8	Notification	Email confirmation sent to the team leader after successful insert or update.
9	Transform History & Import Log	Reports inserted, updated, skipped and error counts for audit and troubleshooting.

Data Flow

1. The admin prepares the spreadsheet with columns such as Team ID, Team Size, Team Leader and Team Member names.
2. The file is uploaded through System Import Sets > Load Data, creating a Data Source and loading rows into the Import Set staging table.
3. A Transform Map matches spreadsheet columns to Team Enrollment fields and coalesces on Team ID, so existing teams are updated instead of duplicated.
4. Transform scripts and business rules validate every row; invalid rows are skipped and logged.
5. Valid rows are inserted or updated in the Team Enrollment table.
6. A notification email confirms enrolment to the team leader, and the Transform History records the results.

Technology Stack

Layer	Technology
Platform	ServiceNow (cloud-based, Now Platform)
Import features	Import Sets, Data Sources, Transform Maps, Field Maps, Coalesce
Data format	Microsoft Excel (.xlsx) / CSV
Automation	Business Rules, Transform Scripts (JavaScript), Email Notifications
Monitoring	Transform History, Import Set Rows, System Logs

Team Details

Role	Name
Team Leader	SRIKAVI T
Team Member	ISWARYA S
Team Member	YOGASRI S
Team Member	KEERTHIKA L

Team ID: SWTID-2026-8331 | Team Size: 4 | Project files: https://drive.google.com/drive/folders/1YpCUwkQuALCmuGVehWFcmxaglr-0dZCD